

Handling Missing Data in ETL

Q1. What are the most common reasons for missing data in ETL pipelines?

Ans.

1. Source system errors or incomplete data entry by users.
2. System migration where some fields get lost during transfer.
3. Optional fields that users leave blank intentionally.
4. Network failures during data transfer between systems.

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

Ans.

Blindly deleting rows causes loss of useful data. Other columns in that row may still have important information. If many rows are deleted, the analysis becomes biased and inaccurate. It is better to handle missing values using imputation or flagging instead of deletion.

Q3. Explain the difference between Listwise deletion and Column deletion. Also mention one scenario where each is appropriate.

Ans.

Listwise Deletion:

Removes the entire row when any value in that row is missing.

Scenario - Use when only very few rows have missing values.

Column Deletion:

Removes the entire column when most of its values are missing.

Scenario - Use when a column has 70-80% missing values and is not needed for analysis.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

Ans.

Mean gets affected by extreme values like very high incomes, which makes it unrealistic.

Median is the middle value and is not affected by outliers. So median gives a more accurate replacement for missing income values in skewed data.

Q5. What is forward fill and in what type of dataset is it most useful?

Ans.

Forward fill replaces a missing value with the value from the previous row above it. It is most useful in time series datasets like daily sales, stock prices, or temperature readings where data changes gradually over time.

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

Ans.

After imputation we lose track of which values were originally missing. Creating a flag column before imputation keeps a record of missing data. This helps in data quality analysis and allows models to treat originally missing values differently.

Q7. Consider a scenario where income is missing for many customers. How can this missingness itself provide business insights?

Ans.

If income is missing for many customers, it may indicate a specific group like students or self-employed people who prefer not to share income. This pattern helps businesses understand customer segments and design targeted offers or marketing strategies for that group.

Q8. Listwise Deletion - Remove all rows where Region is missing.

Ans.

Affected Row:

Customer ID 105 - Amit Verma - Pune - Region is NaN

Dataset after deletion:

101	Rahul Mehta	Mumbai	12000	65000	West
102	Anjali Rao	Bengaluru	NaN	NaN	South
103	Suresh Iyer	Chennai	15000	72000	South
104	Neha Singh	Delhi	NaN	NaN	North
106	Karan Shah	Ahmedabad	NaN	61000	West
107	Pooja Das	Kolkata	14000	NaN	East
108	Riya Kapoor	Jaipur	16000	69000	North

Records lost:

1 record was lost (Customer ID 105 - Amit Verma).

Q9. Imputation - Handle missing values in Monthly_Sales using Forward Fill.

Ans.

Before vs After:

102	Anjali Rao	NaN	-> 12000	(filled from row 101)
104	Neha Singh	NaN	-> 15000	(filled from row 103)
106	Karan Shah	NaN	-> 18000	(filled from row 105)

Why forward fill is suitable:

Sales data is ordered sequentially so the previous value is a reasonable estimate for the missing one.

Q10. Flagging Missing Data - Create a flag column for missing Income.

Ans.

Income_Missing_Flag (0 = present, 1 = missing):

101	Rahul Mehta	65000	0
102	Anjali Rao	NaN	1
103	Suresh Iyer	72000	0
104	Neha Singh	NaN	1
105	Amit Verma	58000	0
106	Karan Shah	61000	0
107	Pooja Das	NaN	1
108	Riya Kapoor	69000	0

Total missing:

3 customers have missing income - Anjali Rao, Neha Singh, Pooja Das.